TASK 1: HANGMAN GAME

Complete Python Mini-Project

Goal: Create a simple text-based Hangman game in which a player guesses a hidden word one letter at a time.

Simplified scope

• Use a predefined list of five different words.

• Allow a maximum of six incorrect guesses.

• Use console input and output only (no graphics or audio).

• Choose the secret word randomly.

1. Python Source Code

```
import random

# Five predefined words
words = ["apple", "tiger", "python", "school", "computer"]

# Select one word randomly
word = random.choice(words)
guessed_letters = []
wrong_guesses = 0
max_guesses = 6

print("Welcome to the Hangman Game!")
print("Guess the hidden word one letter at a time.")

while wrong_guesses < max_guesses:
    # Display letters guessed correctly and blanks for others
    display_word = ""
    for letter in word:
        if letter in guessed_letters:
            display_word += letter + " "
        else:
            display_word += "_ "

    print("\nWord:", display_word)
    print("Incorrect guesses remaining:", max_guesses - wrong_guesses)

    # Check whether every letter has been guessed
    if all(letter in guessed_letters for letter in word):
        print("Congratulations! You guessed the word:", word)
        break

    guess = input("Guess one letter: ").lower().strip()

    # Validate input
    if len(guess) != 1 or not guess.isalpha():
        print("Please enter exactly one alphabet letter.")
        continue

    if guess in guessed_letters:
        print("You already guessed that letter. Try another.")
        continue

    guessed_letters.append(guess)

    if guess in word:
        print("Correct guess!")
    else:
        wrong_guesses += 1
        print("Incorrect guess!")

# If the player uses all six incorrect guesses, the game ends
if wrong_guesses == max_guesses:
    print("\nGame over! The correct word was:", word)
```

```
print("Thank you for playing!")
```

2. How the Program Works

Step 1 – Import random: The random module helps the program choose a word from the list.

Step 2 – Create the word list: Five predefined words are stored in a Python list.

Step 3 – Choose a secret word: random.choice(words) selects one word at random.

Step 4 – Start the loop: A while loop continues while the player has fewer than six incorrect guesses.

Step 5 – Display the word: Correctly guessed letters appear; letters not yet guessed are shown as underscores.

Step 6 – Read and validate input: The program accepts one alphabet letter and rejects invalid or repeated guesses.

Step 7 – Check the guess: A correct letter is accepted. An incorrect letter increases the wrong-guess count.

Step 8 – Finish the game: The player wins by guessing all letters or loses after six incorrect guesses.

3. Key Concepts Used

Concept	Use in the program
random module	Selects a random word from the predefined list.
while loop	Repeats turns until the player wins or reaches the guess limit.
if-else conditions	Checks whether a guess is correct, invalid, or repeated.
strings	Store the secret word and the player's letter input.
lists	Store available words and letters already guessed.
for loop	Checks each letter to build the displayed word.
input() / print()	Read the player's guess and show game messages.
all()	Checks whether every letter in the word has been guessed.

4. Sample Output

```
Welcome to the Hangman Game!
Guess the hidden word one letter at a time.

Word: _ _ _ _ _
Incorrect guesses remaining: 6
Guess one letter: a
Correct guess!

Word: a _ _ _ _
Incorrect guesses remaining: 6
Guess one letter: z
Incorrect guess!

Word: a _ _ _ _
Incorrect guesses remaining: 5
...
```

Note: The actual word and output will vary because the program chooses a word randomly.

5. Algorithm

1. Start the program.

2. Create a list of five words and randomly choose one secret word.

3. Set the guessed-letter list to empty and incorrect guesses to zero.

4. Display the current word with underscores for unguessed letters.

5. Ask the player to enter one letter and validate it.

6. If the letter is correct, display it; otherwise increase the incorrect-guess count.

7. Repeat until all letters are guessed or six incorrect guesses are made.

8. Display the result and stop.

6. Conclusion

The Hangman Game is a beginner-friendly Python project. It demonstrates random selection, loops, conditional statements, strings, lists, input validation, and console output. The player wins by guessing the secret word before reaching six incorrect guesses.